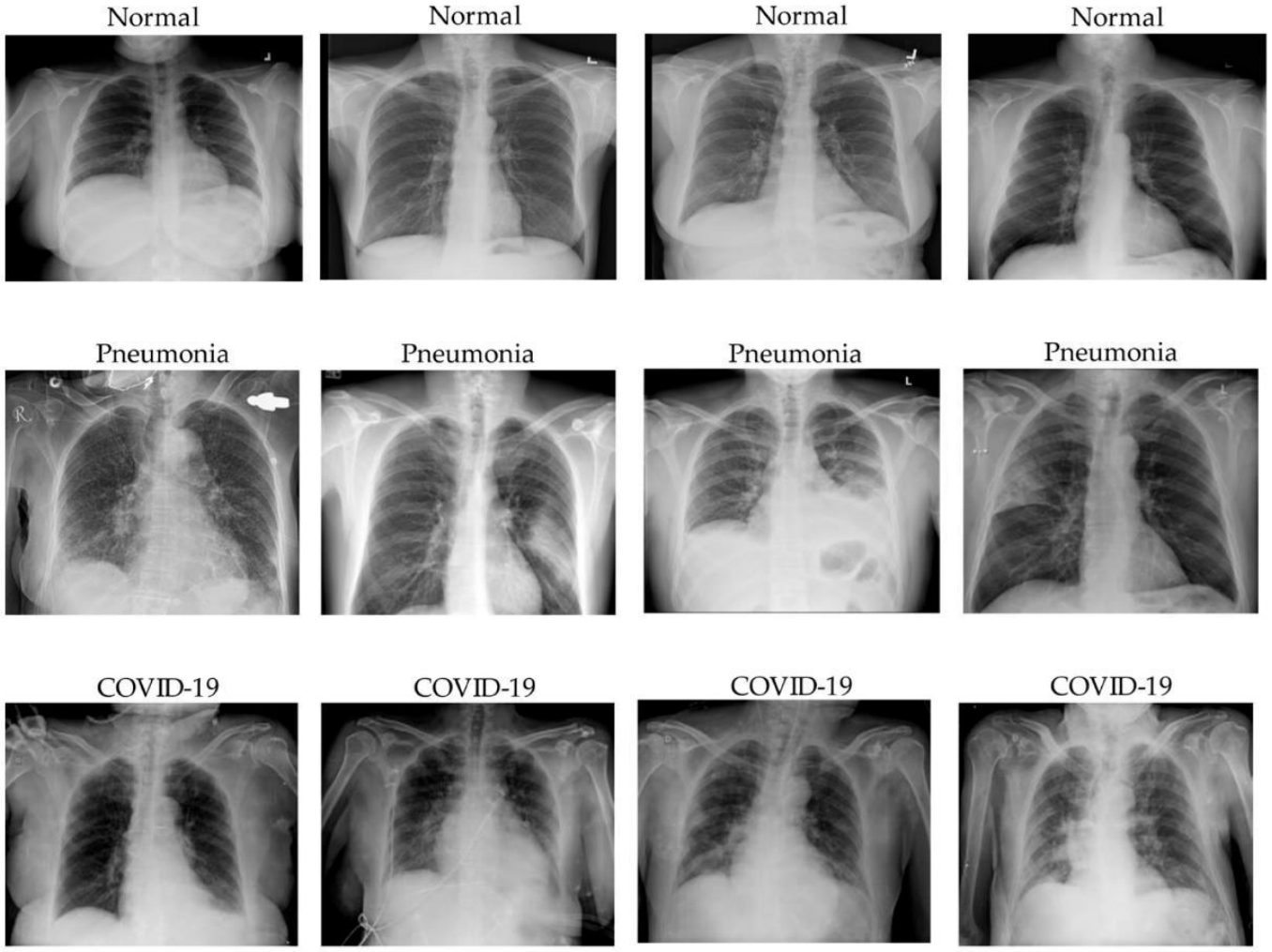
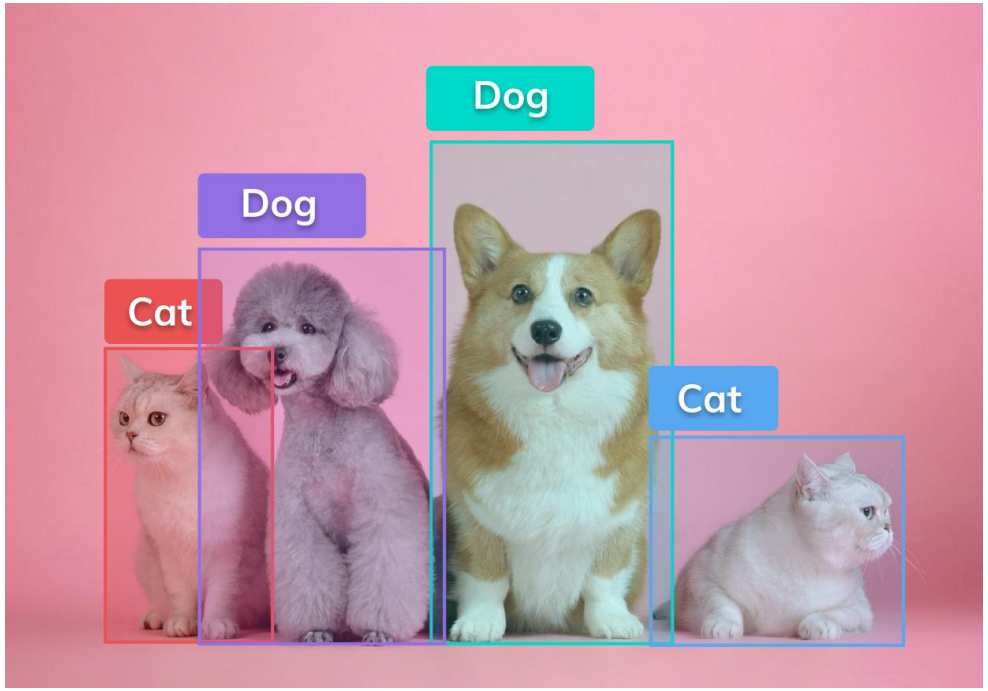
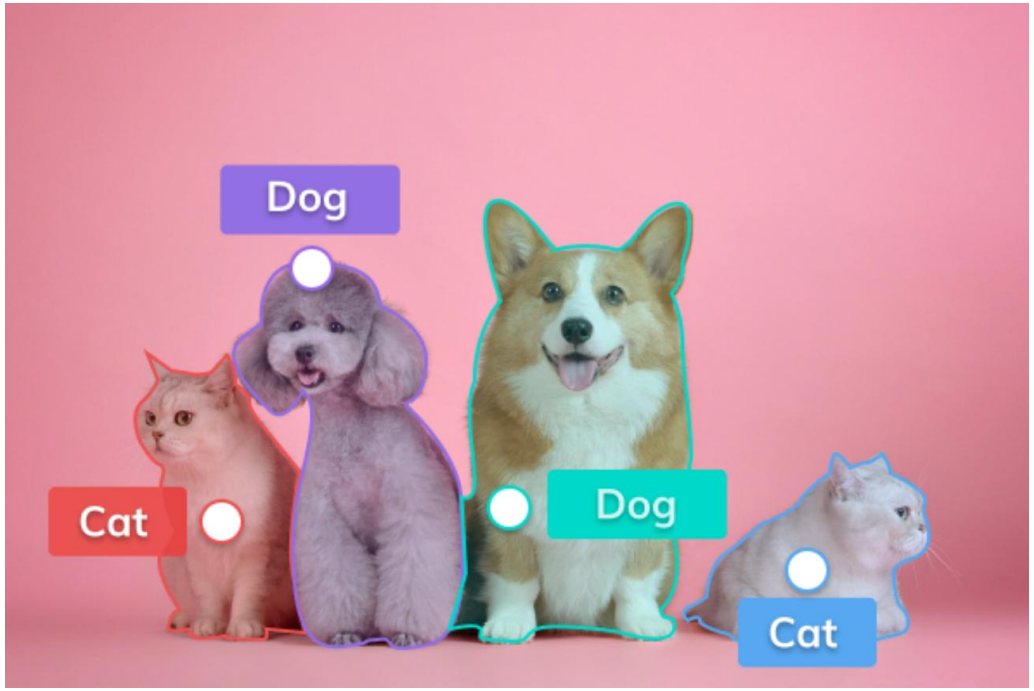
There are various prediction tasks in computer vision: Classification



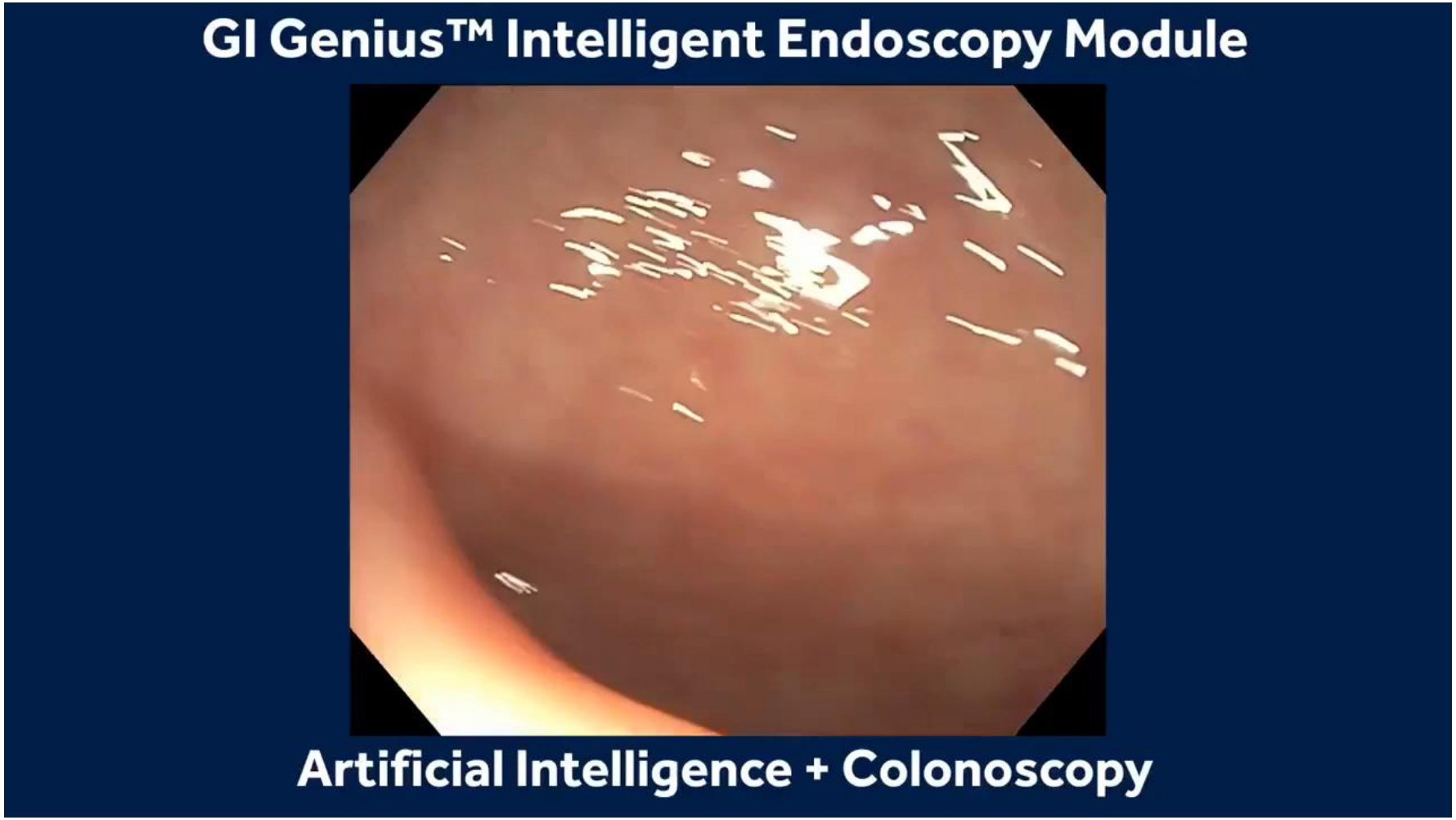
Le Dinh, T, et al. COVID-19 Chest X-ray Classification and Severity Assessment Using Convolutional and Transformer Neural Networks. Appl. Sci. 2022, 12, 4861

There are various prediction tasks in computer vision: Detection & segmentation

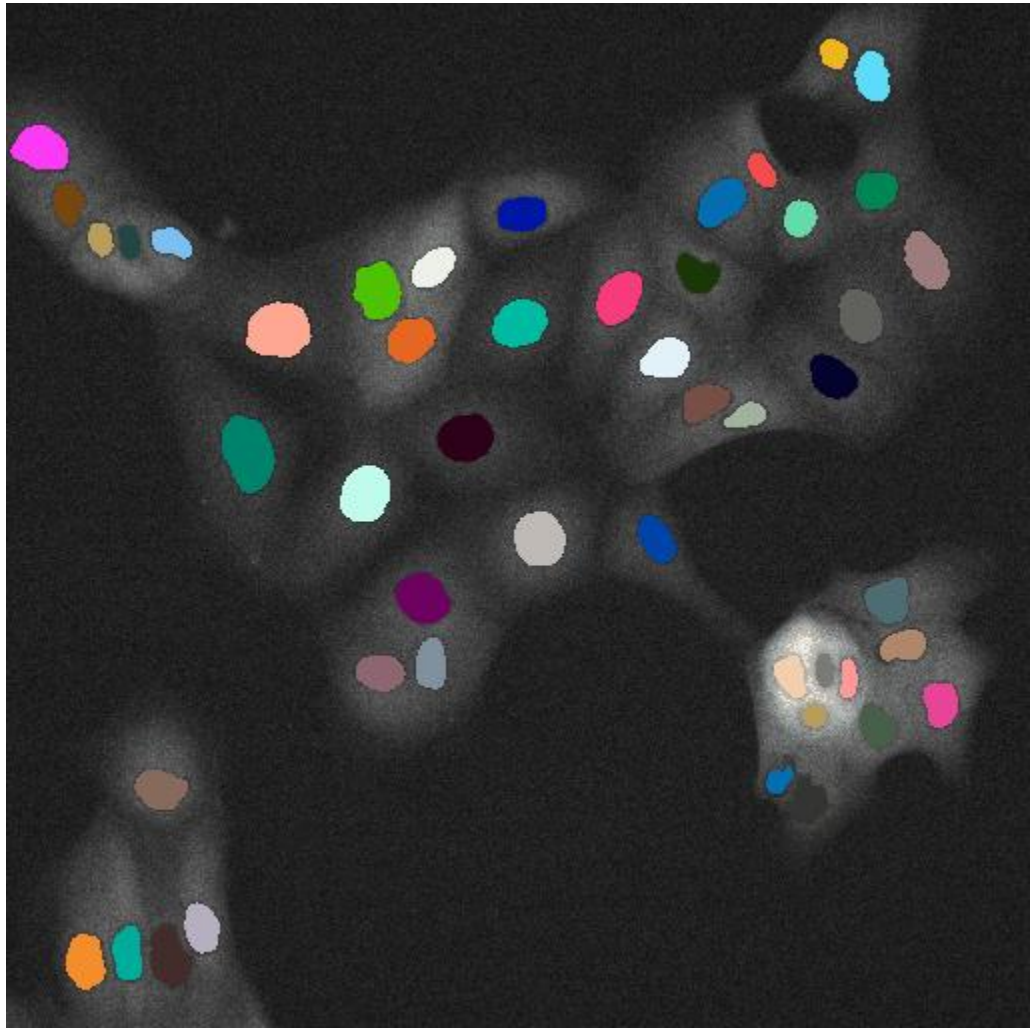




There are various prediction tasks in computer vision: Object detection

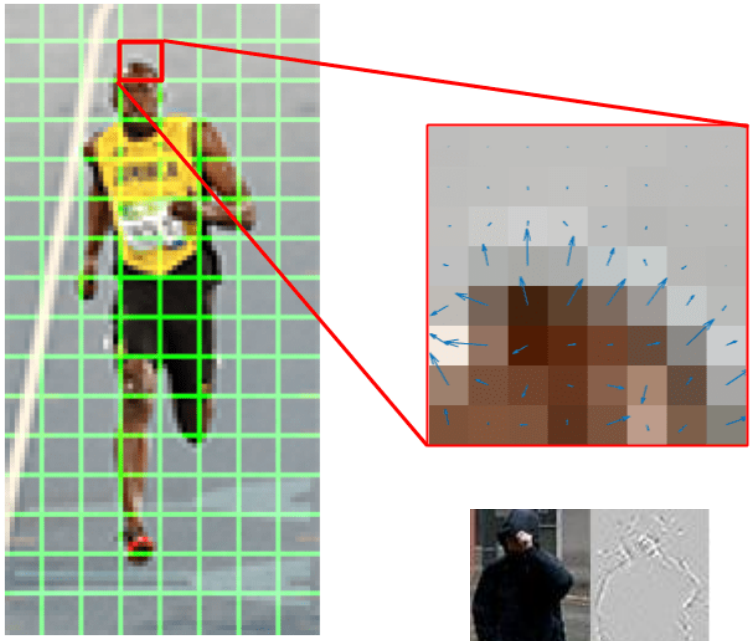


There are various prediction tasks in computer vision: Object tracking



https://imaging.cs.msu.ru/en/research/cell-tracking

Methods before CNNs involved manual feature extraction



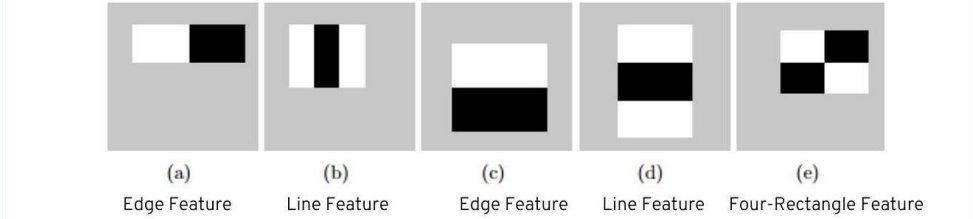
2	3	4	4	3	4	2	2
5	11	17	13	7	9	3	4
11	21	23	27	22	17	4	6
23	99	165	135	85	32	26	2
91	155	133	136	144	152	57	28
98	196	76	38	26	60	170	51
165	60	60	27	77	85	43	136
71	13	34	23	108	27	48	110

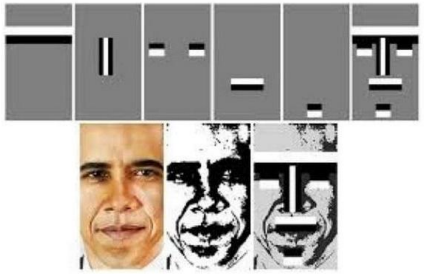
Gradient Magnitude

80	36	5	10	0	64	90	73
37	9	9	179	78	27	169	166
87	136	173	39	102	163	152	176
76	13	1	168	159	22	125	143
120	70	14	150	145	144	145	143
58	86	119	98	100	101	133	113
30	65	157	75	78	165	145	124
11	170	91	4	110	17	133	110

Gradient Direction

Histogram of oriented gradients



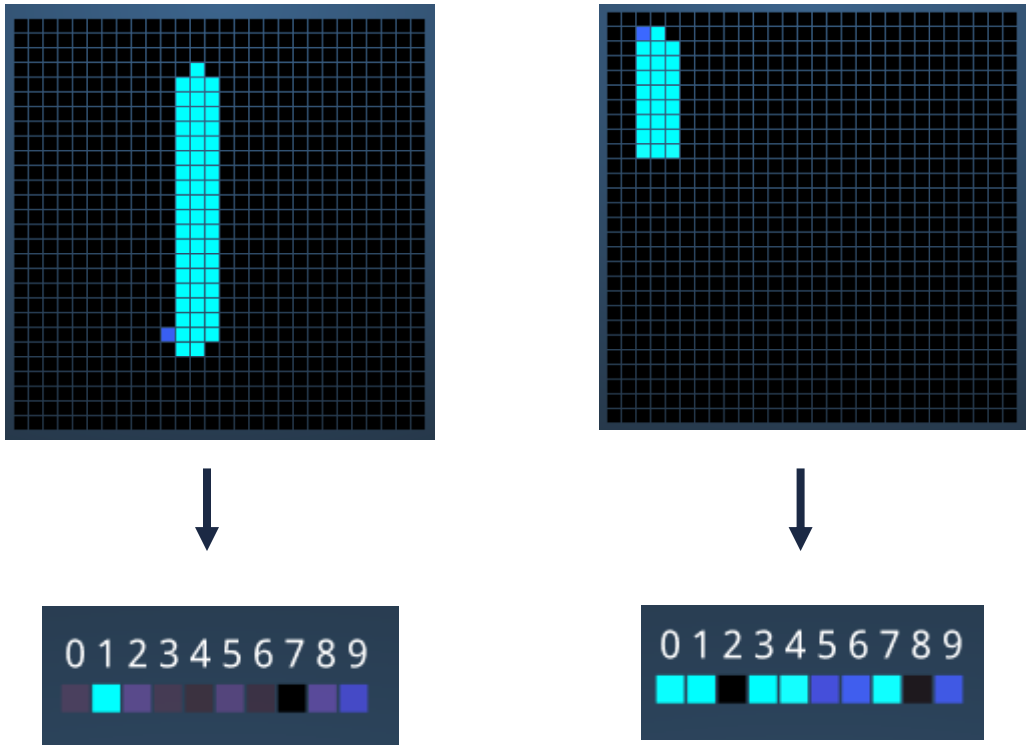




Edge detection

What if neural networks could learn visual patterns instead?

Regular neural network



When digit is centered: good prediction

When digit is NOT centered: bad prediction

✅ Can learn visual patterns itself!

❌ Expects the image to stay in the same location (not robust against translation invariance).

https://adamharley.com/nn_vis/mlp/2d.html

Image recognition must be robust against variances

Translation Invariance



Rotation/Viewpoint Invariance



Size Invariance

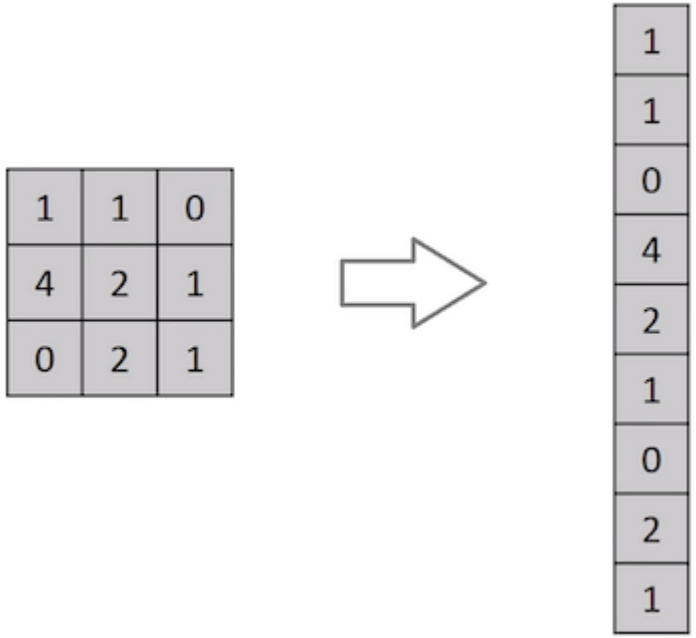


Illumination Invariance



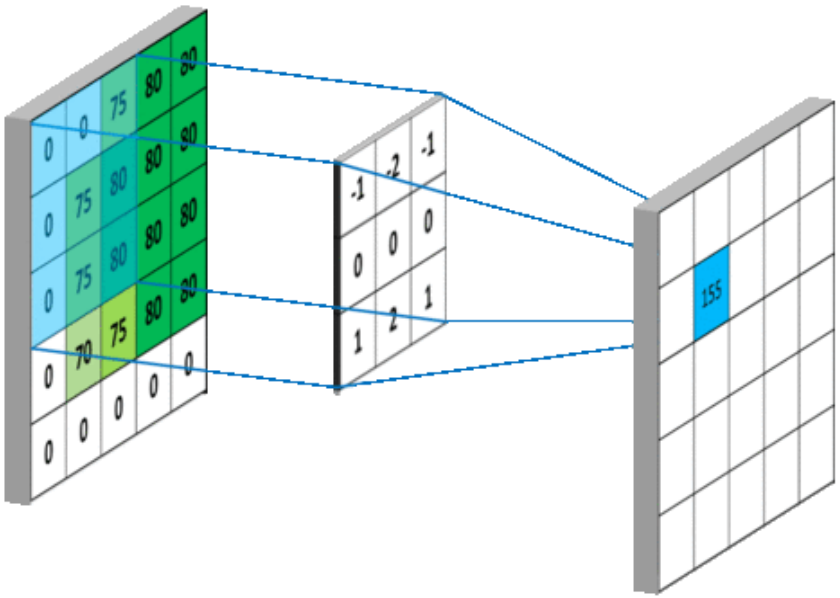
Matt Krause
mattkrau.se

A regular neural network requires flattening of the 2D image, rendering the model not robust against translation invariance

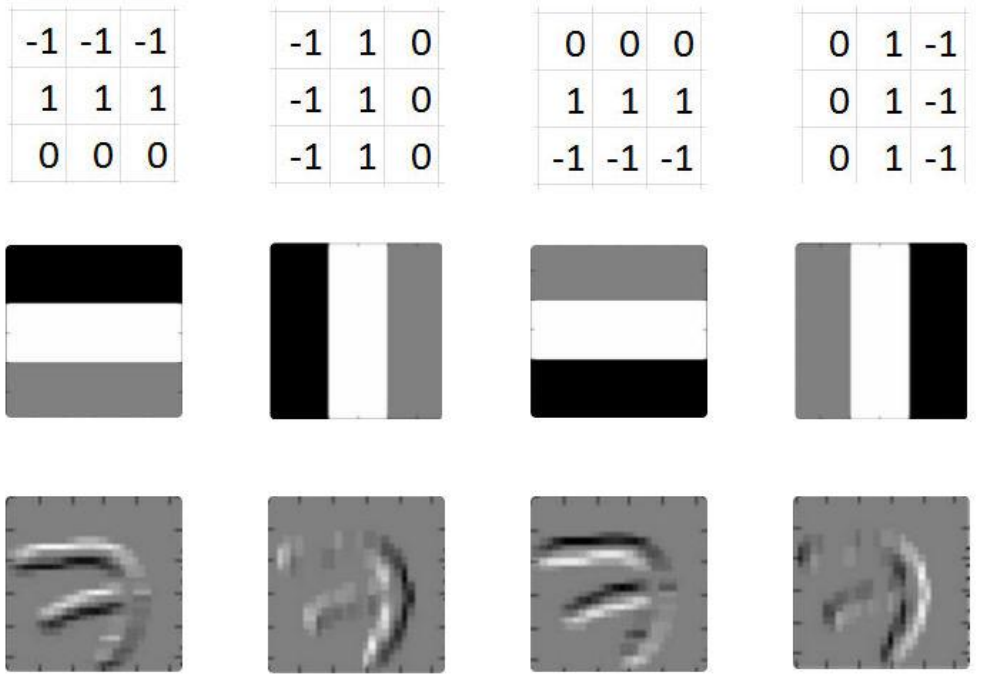


Kernels (aka filters) solve the problem of translation invariance

Kernels ‘convolve’ over the image



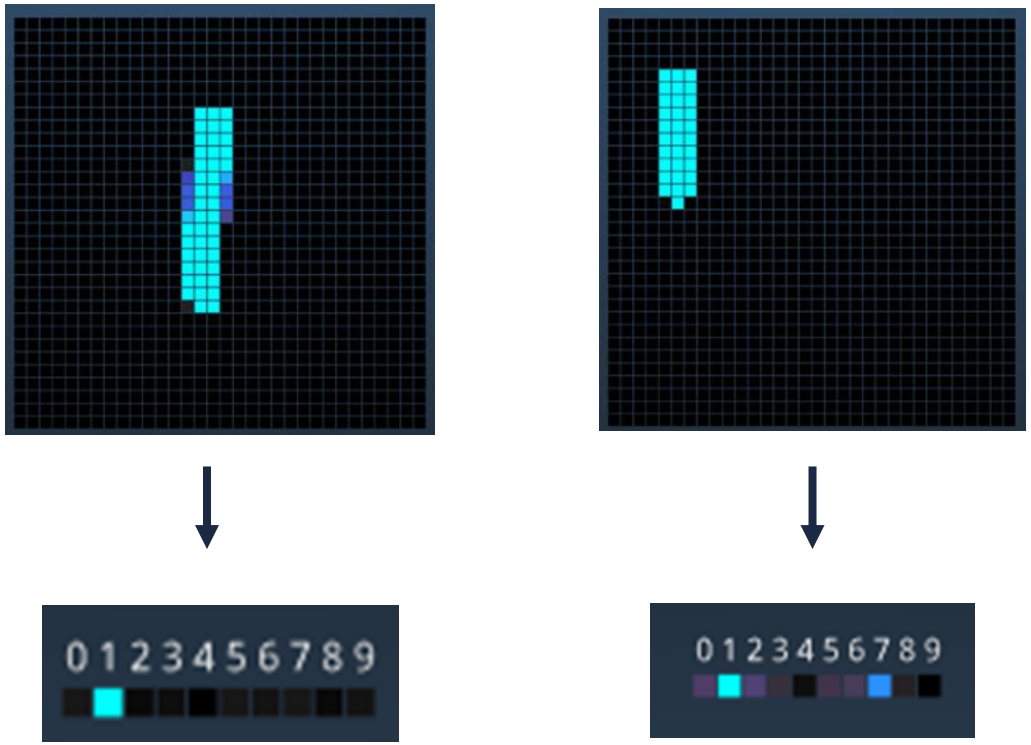
Have been used for manual feature engineering, for instance, for edge detection



Manual edge detection figure: https://octopusinvitro.gitlab.io/blog/machine-learning/convolutional-networks

Kernels (aka filters) solve the problem of translation invariance

Convolutional neural network

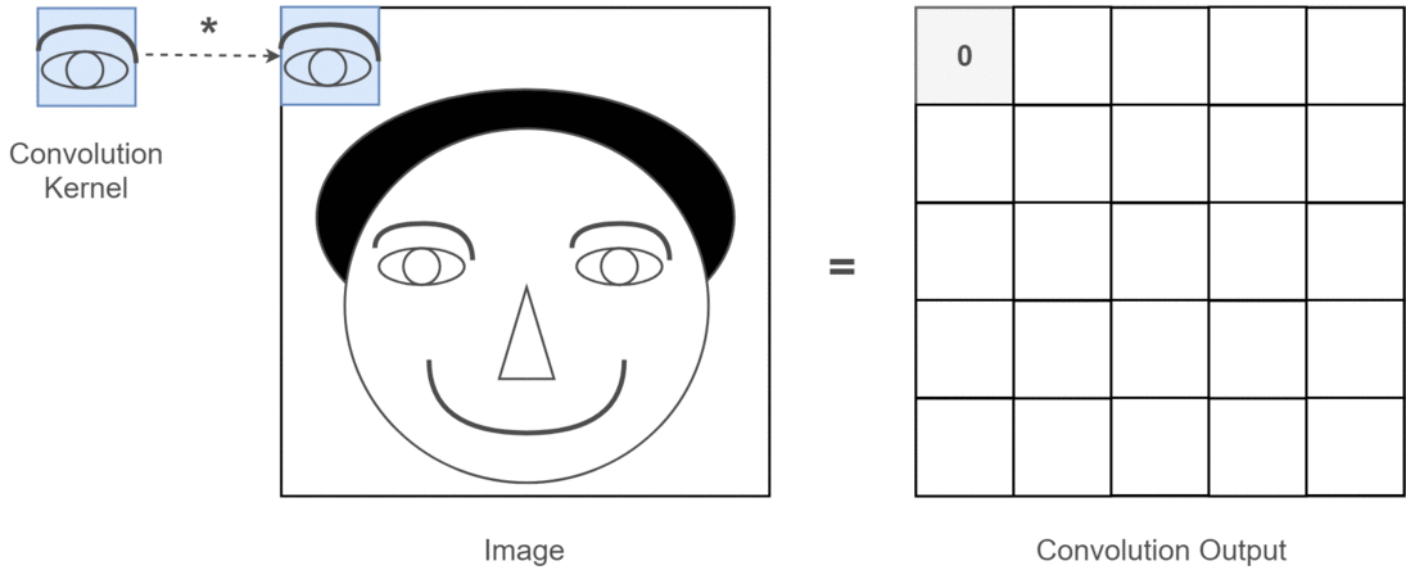


Good prediction independently of object location

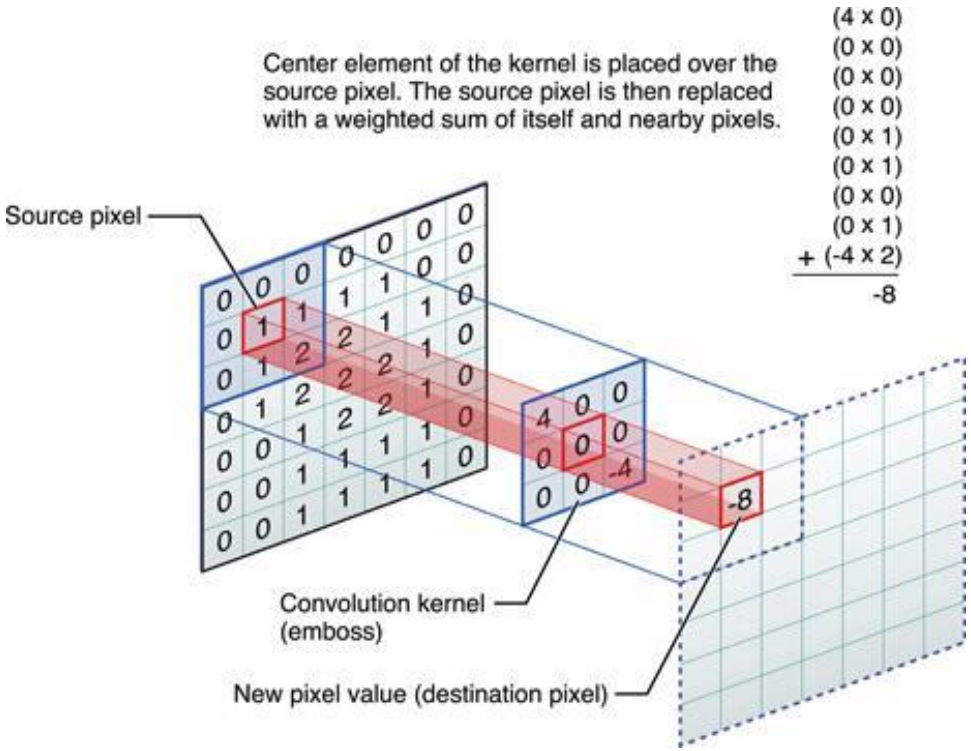
- ✅ Can learn visual patterns itself!
- ✅ Is robust against translation invariance

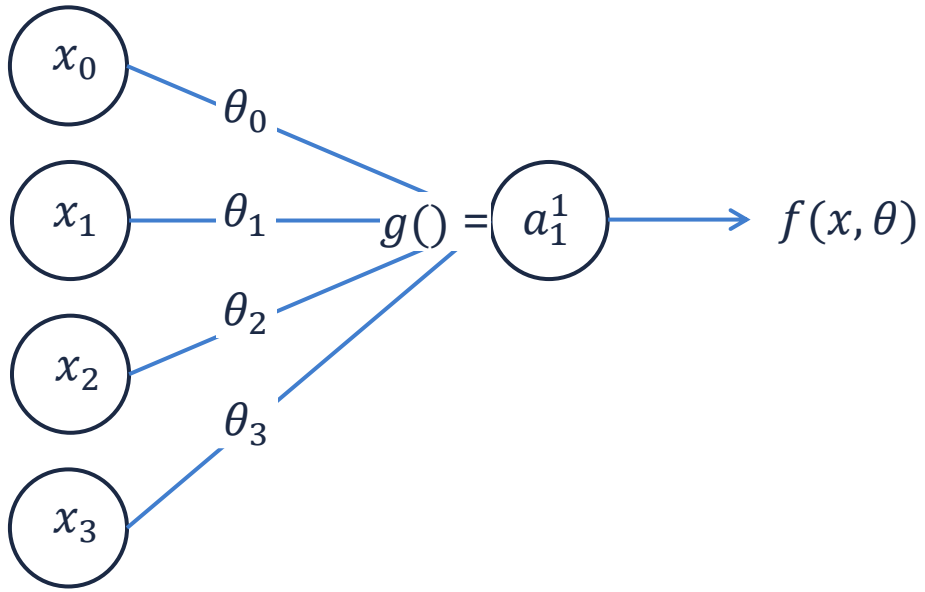
https://adamharley.com/nn_vis/cnn/2d.html

Kernels, simplified



In CNNs, kernels values are learnable parameters of the model





Multiple convolutions can be stacked in layers

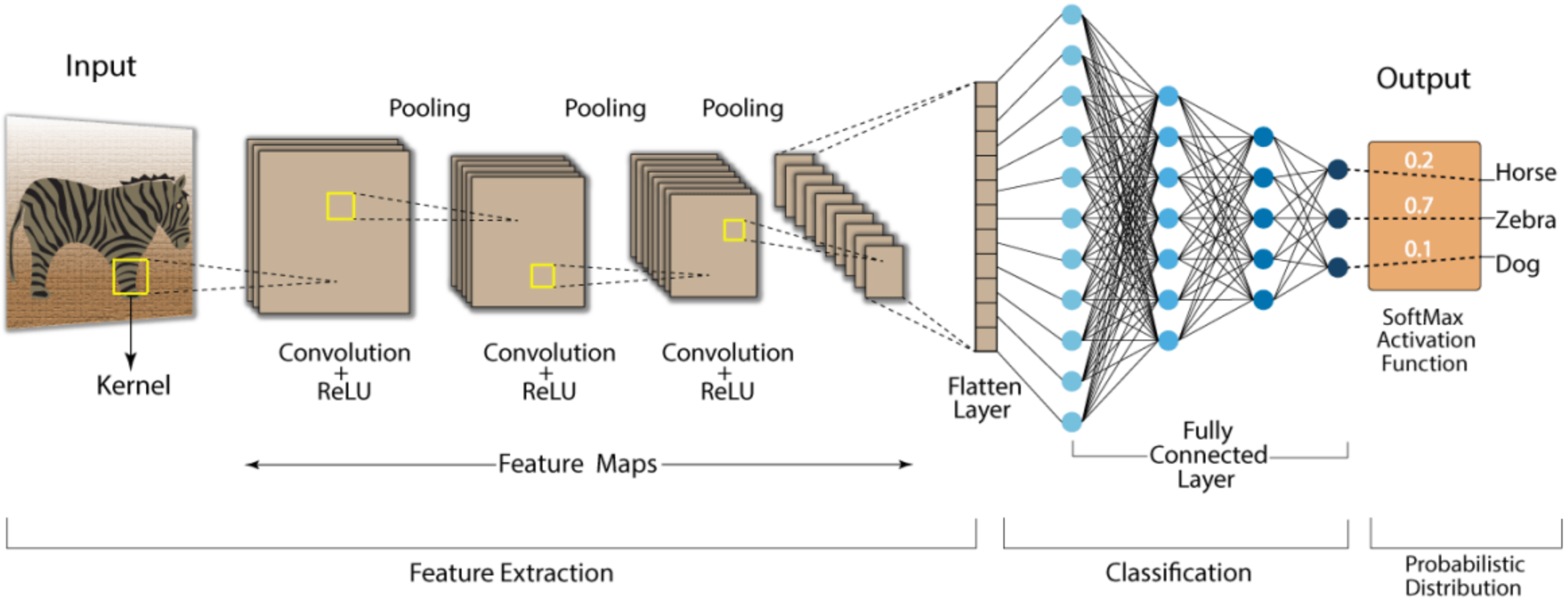
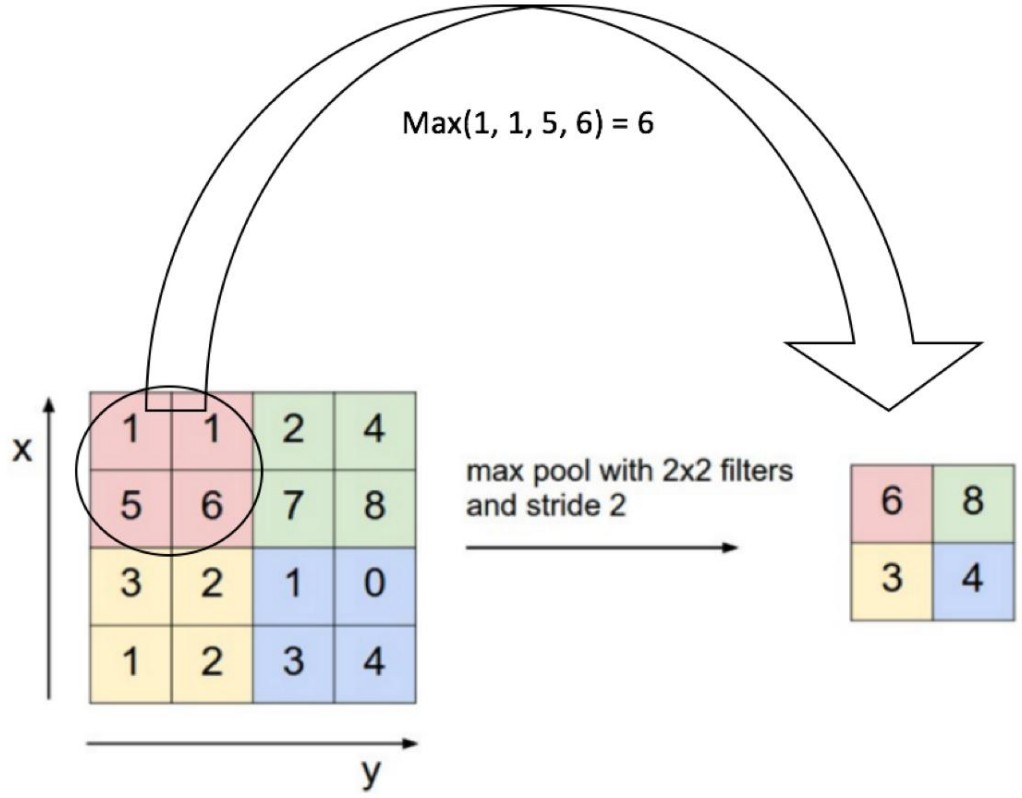
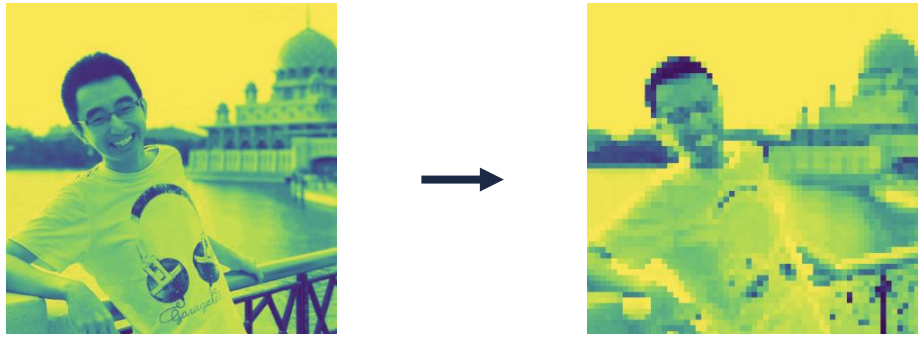


Image from developersbreach.com

Pooling operations downscale the feature map (or image)

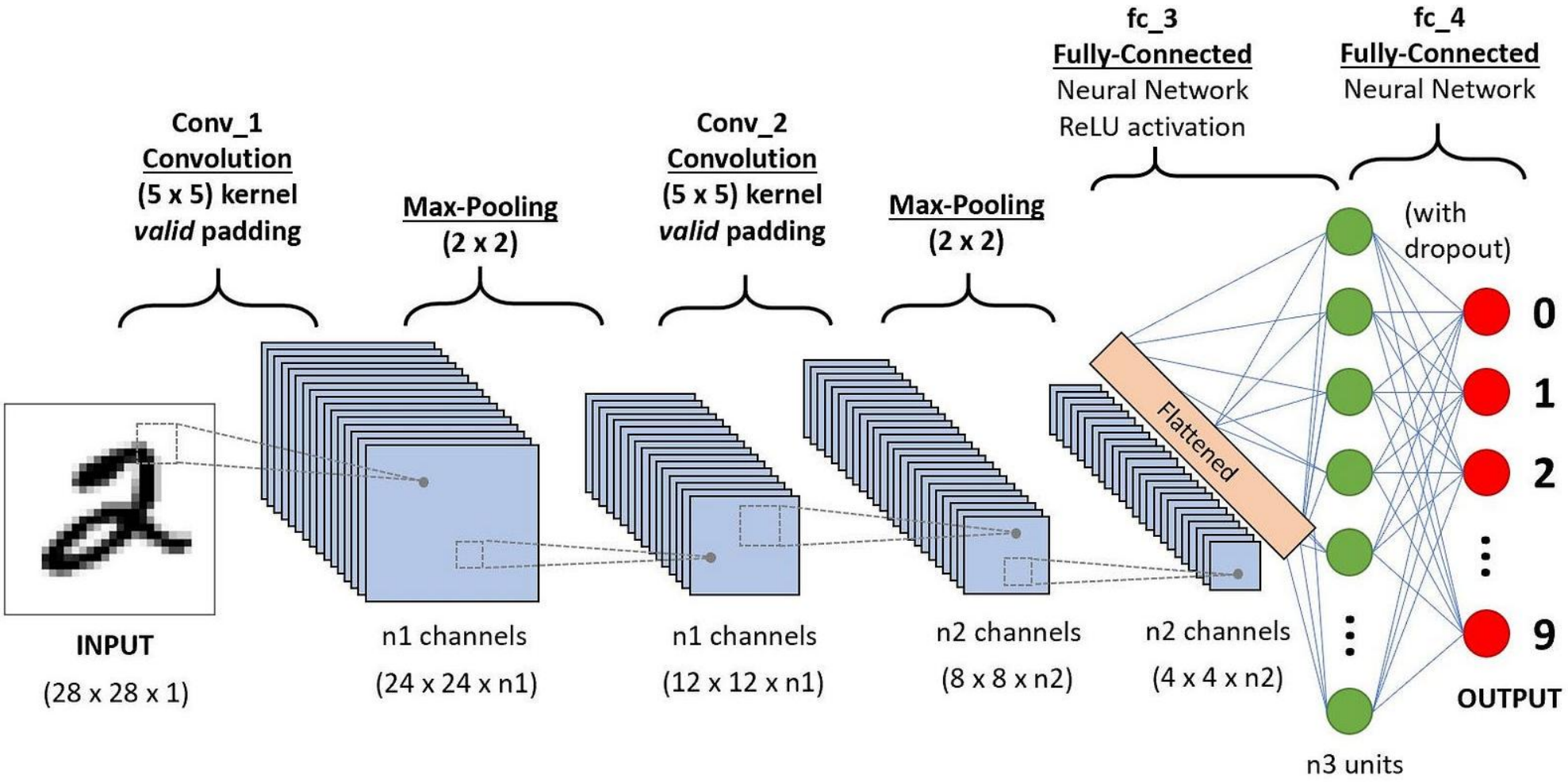




original image

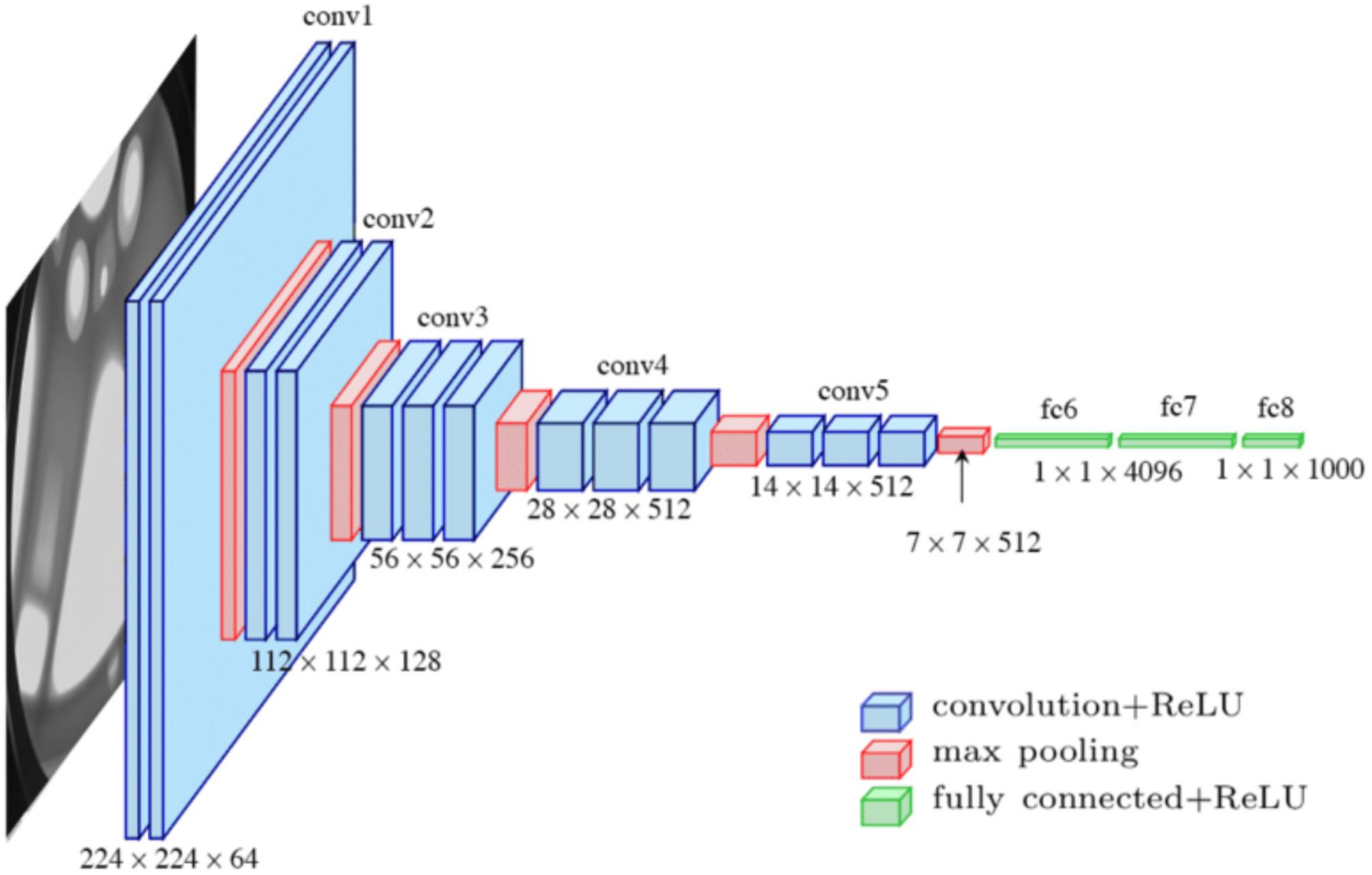
after max pooling

Some noteworthy architectures throughout the years…
LeNet introduced end-to-end CNN-based image recognition



Y. LeCun, et al. Backpropagation Applied to Handwritten Zip Code Recognition. Neural Comput (1998) doi:10.1162/neco.1989.1.4.541

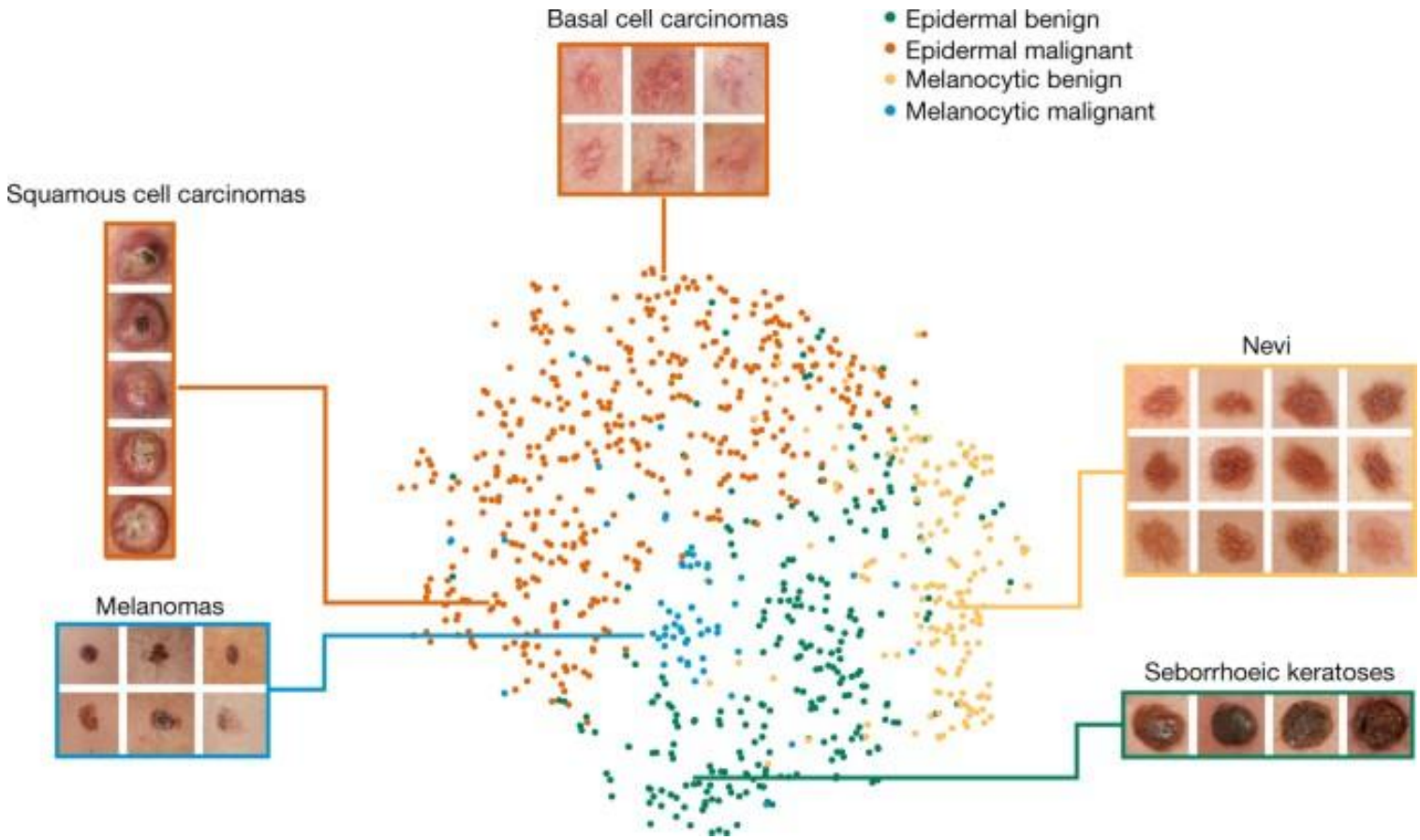
Some noteworthy architectures throughout the years…
VGG showed that deeper CNNs greatly improved accuracy



K Simonyan & A Zisserman. Very Deep Convolutional Networks for Large-Scale Image Recognition. arXiv (2014) doi:10.48550/arXiv.1409.1556

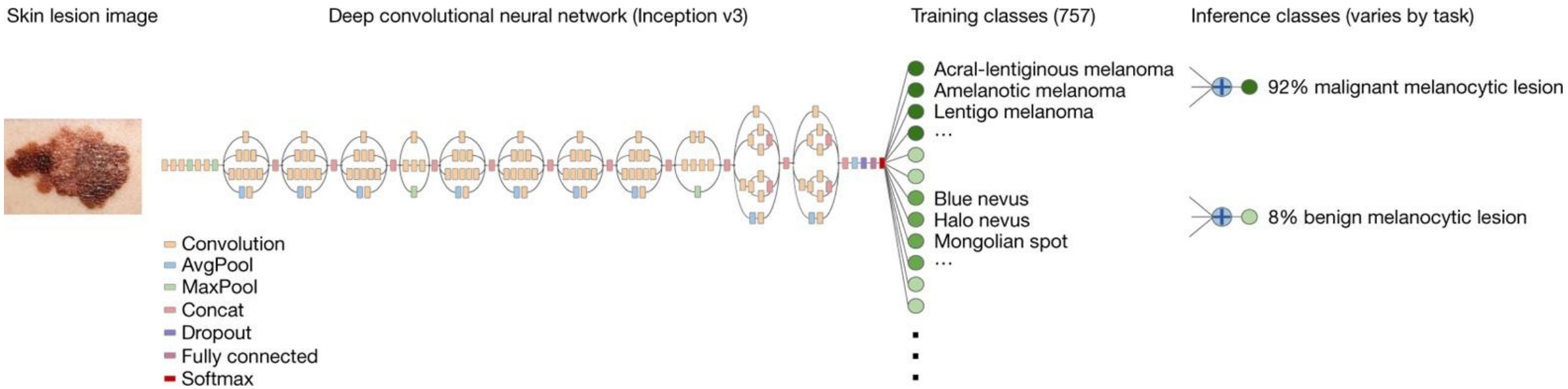
Our dataset for this afternoon: Skin lesions with CNNs
One of the first CNNs that reached expert-level performance





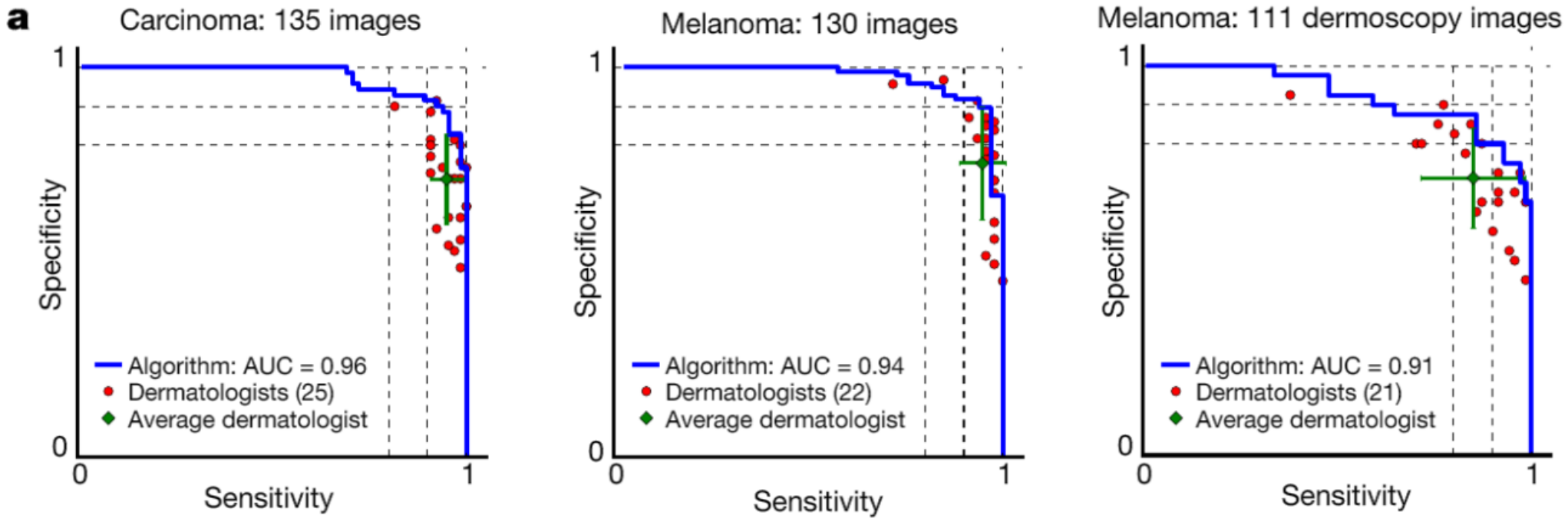
Esteva et al., Dermatologist-level classification of skin cancer with deep neural networks, Nature (2017)

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Esteva et al., Dermatologist-level classification of skin cancer with deep neural networks, Nature (2017)